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TEST CASE REPORT

TEST CASE REPORT – BLACK-BOX (TC-01 to TC-05)

GENERAL INFORMATION	
Test Stage:	☒ Unit ☒ Functionality ☐ Integration ☐ System ☐ Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot
Test Date:	9/19/2026 System Date, if applicable: N/A
Tester:	Student (Author) Test Case Number: TC-01 to TC-05
Test Case Description:	Verifies the Employee, Payroll & Reporting functions of the system via black-box testing: adding/removing employees, calculating salary, and generating reports.
Results:	☒ Pass ☐ Fail Incident Number, if applicable: N/A
INTRODUCTION	
Requirement(s) to be tested:	FR-01: add/update/remove employee records. FR-02: calculate and process employee salary payments. FR-03: generate employee and payment reports. FR-04: validate employee data (ID, salary ≥ 0) before processing.
Roles and Responsibilities:	Student – test designer, tester and reporter: writes test cases, executes them against the Java implementation, and records results.
Set Up Procedures:	Instantiate PayrollSystem in a Java test class; load sample employee records (e.g., Alice, Bob) before each run.
Stop Procedures:	Clear the employee list and dispose of the PayrollSystem instance after the test run completes.
ENVIRONMENTAL NEEDS	
Hardware:	Standard PC (any OS) capable of running the JVM.
Software:	JDK 17 (or later) and a Java IDE / JUnit runner. Interacts only with the PayrollSystem and Employee classes.
Procedural Requirements:	None – tests are run independently in isolation, in the order listed below.
TEST	
Test Items and Features:	addEmployee(), removeEmployee(), calculateSalary(), generateReport() – FR-01–FR-04.
Input Specifications:	TC-01: addEmployee(1, "Alice", 500000) TC-02: calculateSalary(Alice) TC-03: addEmployee(2, "Bob", -100) TC-04: generateReport() with 2 employees TC-05: removeEmployee(1)
Procedural Steps:	Run TC-01–TC-05 by calling each public method listed with its input and observing the return value/state.
Expected Results of Case:	TC-01: employee added TC-02: returns 500000 TC-03: rejected with validation error TC-04: report lists both employees and totals TC-05: employee removed
ACTUAL RESULTS	
Output Specifications:	TC-01–TC-05 all passed – employee list, salary value, and report contents matched the expected results in every case. Overall result: PASS (5/5).

TEST CASE REPORT – WHITE-BOX (TC-06 to TC-09)

GENERAL INFORMATION	
Test Stage:	☒ Unit ☒ Functionality ☐ Integration ☐ System ☐ Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot
Test Date:	9/19/2026 System Date, if applicable: N/A
Tester:	Student (Author) Test Case Number: TC-06 to TC-09
Test Case Description:	Verifies the internal branches of calculateSalary() via white-box testing to achieve full branch coverage.
Results:	☒ Pass ☐ Fail Incident Number, if applicable: N/A
INTRODUCTION	
Requirement(s) to be tested:	FR-02: calculate and process employee salary payments. FR-04: validate employee data (ID, salary ≥ 0) before processing.
Roles and Responsibilities:	Student – test designer, tester and reporter: writes test cases, executes them against the Java implementation, and records results.
Set Up Procedures:	Instantiate PayrollSystem in a Java test class; load sample employee records (e.g., Alice, Bob) before each run.
Stop Procedures:	Clear the employee list and dispose of the PayrollSystem instance after the test run completes.
ENVIRONMENTAL NEEDS	
Hardware:	Standard PC (any OS) capable of running the JVM.
Software:	JDK 17 (or later) and a Java IDE / JUnit runner. Interacts only with the PayrollSystem and Employee classes.
Procedural Requirements:	None – tests are run independently in isolation, in the order listed below.
TEST	
Test Items and Features:	Internal branches of calculateSalary() – FR-02, FR-04.
Input Specifications:	TC-06: salary = 500000 (valid) TC-07: salary = -1 (invalid) TC-08: employee = null TC-09: salary = 0 (boundary)
Procedural Steps:	Run TC-06–TC-09 by calling calculateSalary() directly with each input, tracing the internal branch executed, to achieve full branch coverage.
Expected Results of Case:	TC-06: returns 500000, no exception TC-07: throws IllegalArgumentException TC-08: throws NullPointerException TC-09: returns 0
ACTUAL RESULTS	
Output Specifications:	TC-06–TC-09 all passed – all four branches of calculateSalary() (valid, negative, null, boundary) executed and returned/threw exactly as expected, giving 100% branch coverage. Overall result: PASS (4/4).